Math 115A Spring 2026 Discussion 2 Worksheet

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1 Introduction

Last time, we got acquainted with three of the most common types of proof - direct proofs, proofs by contrapositive, and proofs by contradiction - and used them to proof some results in number theory. We have a few routes we can go down today (all of which will be included in this handout); these are:

1. Proofs by induction
2. Proofs with fields and vector spaces
3. Spotting incorrect proofs

We'll have a vote on what order we'll cover these in (with the implication being we may not actually get to the third topic).

2 Induction

Induction is a powerful technique for proving statements about the natural numbers (and other objects that can be counted). The idea is as follows: we have a statement involving a number n; we show that if it is true for $n-1$, then it is true for n, and if we do this generally enough the same proof will also show that if it is true for $n-2$ it is true fro $n-1$, etc.; then we show that it is true for some small "base case," typically $n = 0$ or $n = 1$. In other words, we show that the statement we wish to prove can always be made simpler, until it becomes so simple that it is obviously true. To put this into more formal language, every induction proof follows three steps (which are typically executed in reverse order to the way I have summarized them above):

Outline of an inductive proof: Suppose we have a family of statements $\{P_n\}$, where n ranges over the natural numbers. (Almost always these statements involve the number n in some way). To prove that P_n is true for all $n \geq 1$, it is enough to do the following:

(a) First, show that P_1 is true.

(b) Now, *assume* that P_k is true; from this assumption, show that P_{k+1} is true. (This shows that if P_k is true, then so is P_{k+1}).

Exercise: How should we modify this process if we want to show that P_n is true for all $n \geq 3$?

Exercise: Convince yourself that induction works as advertised.

This is a lot to take in in the abstract, so it's best to see a few examples.

Theorem: The sum of the first n natural numbers is $\frac{n(n+1)}{2}$. (Here the "natural numbers" start at 1).

We'll break it into steps, each presented as an exercise.

Step 1: What are the statements P_n? What is the statement P_1?

Step 2: Prove P_1. (This is not hard).

Step 3: Write down P_{n+1} and P_n. By manipulating P_n, show that if it is true then so is P_{n+1}.

Warning: A common error in these sorts of proofs is to manipulate P_{n+1} until it is reduced to P_n. Even many professional mathematicians will do this from time to time! Often this is fine, but sometimes not. See the "invalid proofs" section for an example.

Exercise: Show that the sum of the squares of the first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$.

Challenge: Propose and prove a formula for the sum of the cubes of the first n natural numbers.

It is sometimes useful to use the so-called "strong" induction rather than ordinary induction. Here, instead of assuming that P_k is true and proving P_{k+1} is true, we assume that *all the statements* $P_1, \ldots, P_k$ are true and use them to prove that P_{k+1} is true. (The name is a misnomer; this is exactly the same as regular induction, and it's a nice **exercise** to convince yourself of this). Here is an application, which fills in a hole from a proof on last week's handout.

Theorem: Every natural number greater than 1 has a prime factorization.

To prove this we need to decide on a definition of prime and composite. A number $n > 1$ is *prime* if it is only divisible by 1 and itself; it is composite if this is not the case, i.e. if we can write $n = ab$ where a, b are natural numbers and neither is equal to 1.

Step 1: What are the statements P_n?

Step 2: What is the base case?

Step 3: Now, suppose the statements $P_1, \ldots, P_k$ are all true; show that P_{k+1} is true.

Here's one more for the road. You may have heard of the Four-Color Theorem, which states that any map can be colored using just four colors so that no two adjacent regions have the same color. Here's a related statement, which has an elegant argument using induction:

Theorem: Suppose the plane is divided into regions by straight lines. Show that this map can always be colored by *two* colors so that no two regions sharing an edge have the same color.

This may not seem like a statement amenable to an inductive argument, but it is! Indeed, many theorems are more susceptible to this technique than the first appear - but then again, many are not. So while this is an indispensable

technique to keep in your back pocket, it's best not to try to force it where it doesn't actually fit.

3 Fields and vector spaces

Let's prove some of the statements about vector spaces left unproven in the notes.

Exercise: Let V be a vector space. Show that the zero element 0_V is unique.

Exercise: Let V be a vector space over the field F. Show that for $a \in F$ and $v \in V$, we have $av = 0_V$ iff $a = 0_F$ or $v = 0_V$.

A common sort of problem is to show that a given set has the structure of a vector space. Once you get a feel for these, they're mostly all the same, but it never hurts to have an example or two.

Exercise: Show that $\mathbb{C}$ is a vector space over $\mathbb{R}$ (using the ordinary addition and multiplication operations).

Exercise: More generally, suppose that F is a field and $E \subseteq F$ is a subfield (i.e. a subset of F which is closed under addition and multiplication, and which contains identities and inverses). Show that F is a vector space over E.

Exercise: Let V be the set of functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_{-\infty}^{\infty} f(x)dx$ is finite. Show that V is a vector space over $\mathbb{R}$, with the addition and scalar operations you would expect. You may assume any facts from calculus you might wish to use.[1]

As a reminder, the axioms which any vector space V (over a field F) must satisfy are:

- (VS1): for all $x, y \in V$, $x + y = y + x$.
- (VS2): for all $x, y, z \in V$, $(x + y) + z = x + (y + z)$
- (VS3): There exists $0_V \in V$ such that for all $x \in V$, $0_V + x = x + 0_v = x$.
- (VS4): For each $x \in V$, there exists a corresponding $y \in V$ such that $x + y = 0_V$.[2]
- (VS5): For all $x \in V$, $1_F \cdot x = x$.
- (VS6): For all $a, b \in F$ and all $x \in V$, $(ab)x = a(bx)$
- (VS7): For all $a \in F$ and all $x, y \in V$, $a(x + y) = ax + ay$.
- (VS8): For all $a, b \in F$ and all $x \in V$, $(a + b)x = ax + bx$.

[1] This vector space is very closely related to what is usually called $L^1(\mathbb{R})$, although in order to have the "true" $L^1(\mathbb{R})$ you need a slightly different definition of the integral. A related vector space is $L^2(\mathbb{R})$, which consists of functions f such that $\int_{-\infty}^{\infty}(f(x))^2dx < \infty$. You can try proving that this is a vector space as well, but it's much trickier.

[2] As a warning, this is **not** the same claim as "$\exists y \in V \forall x \in V$ such that $x + y = 0_V$; as an exercise, try to figure out the difference for yourself.

Remember that you must check each of these properties to establish that a given set is indeed a vector space. For a slightly funkier example, consider this:

Exercise: Consider the set X of positive real numbers, with the operation $a \oplus b = ab$; we have a "scalar multiplication" $\otimes$ defined as follows: for $a \in \mathbb{R}$, $b \in X$, set $a \otimes b = b^a$. Show that X, together with the given operations, is a vector space over $\mathbb{R}$. (Note that you'll need to find 0_X yourself).

Challenge: Consider a set A, and let V be the set of subsets of A. Find suitable "addition" and "scalar multiplication" operations so that V is a vector space over $\mathbb{F}_2$, the field of two elements mentioned in the notes.

Finally, when trying to wrap your head around a mathematical definition, the non-examples are often as instructive as the examples, if not more.

Exercise: Consider the set $X = \mathbb{R}^2$ with the "addition" operation $\boxplus$ given by $(a, b) \boxplus (c, d) = (a + c + 1, b + d + 1)$, and with ordinary scalar multiplication, that is, $k(a, b) = (ka, kb)$. Show that these operations on X do *not* give it the structure of a vector space over $\mathbb{R}$.

4 Incorrect Proofs

Your sense of what a correct proof is might be refined by seeing a few examples of invalid proofs. Here is an invalid proof that proves an invalid result.

"Proposition": $-2 = 2$.

Proof. We have

$$\begin{aligned} -2 &= 2 \\ (-2)^2 &= 2^2 \\ 4 &= 4 \end{aligned}$$

which is true, and so the original statement is true as well. $\square$

What's the issue?

A similar mistake might arise in the exercises on induction - specifically the problem that asks you to prove a formula for the sum of the first n natural numbers. If you start by writing

$$1 + 2 + \cdots + k + (k + 1) = \frac{(k + 1)(k + 2)}{2}$$

and then subtract $k + 1$ from both sides, you are actually committing the same fallacy as in the above proof that $-2 = 2$. Fortunately, in this case, the mistake is easy to correct; do you see how?

Even many professional mathematicians will sometimes get careless when writing proofs like this, but they will always check for themselves that the argument itself is actually valid. As a general rule of thumb, if you want to show that equation E implies equation E', the correct thing to do is to write E first, then manipulate it until you get E'. It is not correct to start by writing

E' and proceed by manipulating it until you get the given equation E; if all the manipulations are reversible, however, you will be able to rewrite the "incorrect" argument as a correct one. (Note that if you are trying to prove that equation E' holds, the role of E can be played by any equation known to be true).

Next, here's a proof by induction with a subtle error.

"Theorem": All horses are the same color.

Proof. We work by induction: we will show that for any integer n, and any set H consisting of n horses, all the horses in H are the same color.

First, the base case. Let H be a set consisting of a single horse. This horse is certainly the same color as itself, so the base case holds.

Now, suppose we know the statement is true for all sets of k horses. Let H be a set of $k+1$ horses, labeled $h_1, \ldots, h_{k+1}$. We can form two subsets of H, $H_1 = \{h_1, \ldots, h_k\}$ and $H_2 = \{h_2, \ldots, h_{k+1}\}$. By the inductive hypothesis, all the horses in H_1 are the same color as each other; if we can show that h_{k+1} is the same color as the horses in H_1, we will be done. Well, all the horses in H_2 are the same color, and $h_2 \in H_2$, so h_{k+1} is the same color as h_2. But h_2 is the same color as all the horses in H_1. Thus, all the horses in H are the same color. $\square$

But we know from our own experience that not all horses are the same color. What's wrong with the proof?

You also need to be careful about making unjustified claims. For example

"Theorem": $0.\overline{9} \neq 1$.

Proof. Consider the sequence $0.9, 0.99, 0.999, \ldots$. This sequence clearly converges to $0.\overline{9}$. Since each term in this sequence is less than 1, the limit must be less than 1 as well. So, $0.\overline{9} < 1$, which means it is not equal to 1. $\square$

Finally, here's an incorrect proof of a true statement:

Proposition: 5 is odd.

Proof. If 5 were even, we could write $5 = 2k$ for some integer k. Let $k = 3$. Then $2k = 6 \neq 5$. So 5 is not even, which means it must be odd. $\square$